

Design and Evaluation of Volatility Targeting for Systematic Strategies

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Abstract

This project focuses on the design and evaluation of volatility targeting strategies for systematic investment portfolios. The central challenge is that volatility is not directly observable and can change significantly across market regimes. To address this problem, we develop a modular research framework that separates volatility forecasting from portfolio control.

The framework evaluates multiple volatility estimators, including traditional econometric models, machine learning models, and hybrid models. It also tests different portfolio controllers, including naive scaling, regime switching, drawdown protection, and tail-risk-aware methods. A regime-aware combined strategy is then developed using market regime classification to select strategies dynamically across low, medium, and high volatility environments.

The main objective of the report is to examine whether adaptive volatility targeting can improve risk control, reduce drawdowns, and produce more stable risk-adjusted returns compared with buy-and-hold and naive volatility targeting benchmarks.

Keywords

Volatility Targeting; Risk Management; Regime Switching; Hidden Markov Model; QLIKE; Sharpe Ratio; Maximum Drawdown; Systematic Strategies

Executive Summary

This project studies how systematic strategies can control portfolio risk through volatility targeting. In volatile markets, investors face large drawdowns and unstable risk exposure. Traditional static strategies often fail because they do not adjust effectively when market regimes shift.

Our solution is a modular volatility targeting framework. The framework separates the problem into two major components. First, volatility estimators forecast future risk using historical returns, realized volatility, VIX, and rate information. Second, controllers translate those forecasts into portfolio weights and risk-control decisions.

A key insight from the project is that no single model performs best in all market environments. Some models are better in low-volatility regimes, while others perform better during high-volatility periods. Therefore, we design a regime-aware strategy that classifies the market into low, medium, and high volatility regimes and dynamically selects the most appropriate strategy for each regime.

The results show that regime-aware strategy selection can improve portfolio stability. Compared with buy-and-hold, the combined strategy reduces maximum drawdown significantly while maintaining competitive annualized returns. Compared with naive volatility targeting, the combined strategy provides stronger downside protection and better risk-adjusted performance.

The practical impact of this project is a scalable research framework that can be extended beyond a single asset. Future extensions may include multi-asset portfolios, improved regime detection using macroeconomic signals, and hedging overlays using VIX or VXX futures.

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1. Introduction

1.1 Background

In financial markets, volatility is not merely a statistical property of asset returns—it is a fundamental risk signal that governs investor behaviour, portfolio construction, and the pricing of risk premia. Two well-documented phenomena sit at the heart of volatility targeting as a practical investment strategy: the leverage effect and risk aversion.

1.1.1 The Leverage Effect

The leverage effect, first identified by Black (1976) and subsequently formalized by Christie (1982), describes an empirically robust asymmetry in equity markets: negative returns tend to cause disproportionately large increases in volatility, while positive returns of equivalent magnitude produce far smaller volatility responses. The mechanistic explanation is straightforward—when equity prices fall, the debt-to-equity ratio of a levered firm rises automatically, increasing the financial risk borne by shareholders. This structural change in the firm’s capital stack causes the equity to behave as though it were a more levered claim, thereby amplifying subsequent return volatility.

The implications of the leverage effect for active risk management are significant. During periods of market stress, volatility spikes rapidly and asymmetrically, often catching statically-weighted portfolios severely off-guard. A portfolio that is appropriately sized during calm conditions can become dangerously over-levered within days when volatility regimes shift. This dynamic motivates the need for adaptive, volatility-aware position sizing.

1.1.2 Risk Aversion and the Volatility-Return Relationship

A foundational tenet of modern portfolio theory holds that investors are risk-averse: they require compensation—in the form of higher expected returns—for bearing additional risk. In efficient markets, this trade-off should be broadly consistent across time. However, a substantial body of empirical research dating from the work of Ang, Hodrick, Xing, and Zhang (2006) has demonstrated a persistent anomaly in equity markets: periods of elevated volatility are not reliably associated with commensurately higher subsequent returns. To the contrary, high-volatility regimes in equities have historically coincided with poor or negative risk-adjusted performance.

This apparent breakdown of the risk-reward relationship during volatile episodes can be understood through the lens of time-varying risk aversion. In periods of market turbulence, investor sentiment deteriorates sharply, liquidity conditions tighten, and aggregate risk appetite diminishes—sometimes precipitously. This collective reduction in risk tolerance creates selling pressure that further depresses asset prices, whilst simultaneously inflating

measured volatility. The investor who maintains a full, unhedged exposure during such a regime is therefore taking on substantially more risk than during calm periods, yet the expected return for bearing that elevated risk is not proportionately higher. The risk-reward ratio, in effect, deteriorates precisely when it matters most.

This relationship between high volatility and negative expected equity returns provides the empirical and theoretical justification for volatility targeting as an investment discipline. By systematically reducing exposure when volatility rises—and correspondingly increasing it when volatility subsides—a volatility-targeting strategy seeks to maintain a consistent level of risk contribution over time, thereby avoiding the adverse periods where excess risk is taken without adequate compensation.

1.1.3 Volatility Targeting as a Systematic Response

Volatility targeting translates these theoretical observations into a concrete portfolio management rule. Rather than holding a fixed notional position, the strategy dynamically adjusts exposure so that the expected portfolio volatility remains close to a predefined target level. This approach has several well-documented benefits: it smooths the realized risk profile over time, reduces the severity of drawdowns during volatile regimes, attenuates left-tail risk, and facilitates straightforward risk budgeting for portfolio construction and structured product design.

Volatility targeting is today widely embedded in systematic risk-controlled strategies across the asset management industry. It forms the backbone of many risk-parity frameworks, is standard in constant-proportion portfolio insurance (CPPI) structures, and serves as the underlying mechanism in a broad class of volatility-controlled index products—strategies that are commonplace as the underlyings in structured notes, annuity products, and alternative investment mandates.

Despite its conceptual simplicity, the practical implementation of volatility targeting is non-trivial. The design choices made in constructing such a strategy—how volatility is estimated, how exposure is adjusted, how regime changes are handled—have material consequences for performance, turnover, and risk management efficacy. It is precisely this design problem that the present project addresses.

1.2 Problem Statement

Volatility targeting is widely adopted across systematic, risk-controlled investment strategies, serving as a core mechanism in both active asset management and as the underlying risk framework for structured products. Its appeal lies in its conceptual clarity: by dynamically scaling exposure in response to estimated volatility, a strategy can maintain a relatively stable level of realized risk, smooth drawdown profiles, reduce tail risk exposure, and simplify risk budgeting across multi-asset portfolios.

However, there is no single standardized implementation of volatility targeting, and this absence of consensus is itself the central problem. Practical implementations diverge along two primary axes. First, volatility estimators vary considerably—from classical realized volatility and GARCH-family models to exponentially weighted moving averages and, increasingly, machine learning-based forecasting approaches. Each estimator carries distinct assumptions about volatility dynamics, lag sensitivity, and regime behaviour, all of which propagate directly into the portfolio weight signal. Second, exposure adjustment rules differ materially across practitioners, spanning direct proportional scaling, smoothed adjustments, no-trade banding, and regime-conditional rules. These are not merely technical nuances—they have substantive consequences for risk management efficacy, portfolio turnover, and tail behaviour.

The stakes of poor design are concrete. Unstable or ill-specified exposure mechanisms can produce discontinuous weight adjustments that create sharp swings in portfolio composition. These swings complicate delta hedging for structured product overlays, increase P&L volatility, and can generate transaction costs that materially erode the theoretical benefits of the strategy. In the worst cases, an overreactive controller amplifies the very risk it was designed to suppress.

The problem this project addresses is therefore not whether volatility targeting is a sound concept—its theoretical and empirical foundations are well-established—but rather how to rigorously design, compare, and evaluate a family of volatility-targeting implementations under a common research framework. By constructing a shared pipeline in which multiple estimation models and control mechanisms are assessed against identical data, metrics, and backtesting conditions, this project provides a structured basis for understanding which design choices produce strategies that are not only theoretically sound, but practically implementable, hedgeable, and robust across market regimes.

1.3 Project Scope

This project designs and evaluates a family of volatility-targeting strategies applied to S&P 500 futures, each calibrated to a common annualized volatility target of 10%. The analysis draws on three primary data sources: intraday and daily S&P 500 futures return data from DataBento, VIX index data from DataBento as a forward-looking implied volatility proxy, and U.S. Treasury yield curve data from the Wharton Research Data Services (WRDS) platform as a macroeconomic conditioning signal.

Volatility estimation is addressed through both traditional and advanced methods, encompassing Realized Volatility (RV), Exponentially Weighted Moving Average (EWMA), GARCH-family models, Heterogeneous Autoregressive (HAR) models, and XGBoost-based machine learning forecasters. These estimators feed into a range of exposure control mechanisms, including naive volatility scaling, variance-based scaling, regime-switching controllers, and drawdown-aware control rules. All strategies are evaluated within a walk-forward backtesting framework with strict no-look-ahead-bias adherence, prioritising robustness across

market regimes over optimisation for any single performance statistic.

1.4 Project Objectives

The project is structured around the following measurable outcomes:

- Construct a modular, reproducible research pipeline for volatility estimation, exposure control, and strategy backtesting, built on common data inputs and evaluation metrics.
- Benchmark multiple volatility forecasting models—from classical statistical estimators to machine learning approaches—against one another on forecast accuracy, stability, and practical implementability.
- Quantify the impact of each exposure control mechanism on realized risk, returns, drawdown behaviour, and trading efficiency relative to an unmanaged benchmark.
- Evaluate all strategies systematically across four dimensions: risk-adjusted returns, volatility control accuracy, tail risk and drawdown behaviour, and transaction cost efficiency.
- Produce insight-driven diagnostics that explain the mechanisms driving each strategy's performance and identify characteristic failure modes.
- Provide a rigorous comparative framework ensuring all strategies are assessed on an equal, no-look-ahead-bias basis, enabling fair and reproducible conclusions.

2. Methodology Framework

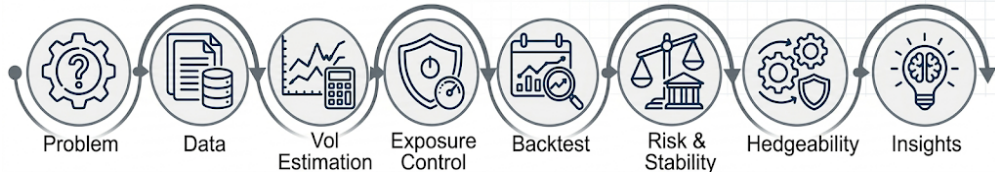


Figure 1: Research Pipeline

2.1 System Overview

The research framework is structured as a modular, end-to-end pipeline that separates the core functional components of a volatility-targeting strategy into distinct, independently testable layers. This architecture enables systematic comparison across models and controllers while ensuring consistent evaluation conditions. The pipeline comprises five principal components:

- **Data Layer** — Ingests, validates, and preprocesses all market data inputs, including return series, volatility proxies, and macroeconomic signals. Constructs the realized volatility benchmark used for model training and evaluation.
- **Volatility Estimation Layer** — Generates daily forward-looking volatility forecasts using each of the implemented models. All models are trained and evaluated on a common benchmark using identical in-sample and out-of-sample windows.
- **Control Layer** — Translates volatility forecasts into daily portfolio weights using predefined exposure control rules. Each controller is designed to balance responsiveness to volatility signals against portfolio stability.
- **Backtesting Engine** — Simulates strategy performance under realistic conditions, incorporating transaction costs, rebalancing constraints, and strict no-look-ahead-bias protocols.
- **Evaluation and Diagnostics Layer** — Produces a structured, multi-dimensional assessment of strategy performance, covering risk-adjusted returns, volatility control accuracy, tail risk, drawdown behaviour, and trading efficiency.

2.1.1 Core Weight Mechanism

The fundamental exposure-setting rule shared across all strategy variants derives the portfolio weight at each time step by scaling the target volatility against the current volatility forecast:

$$w_t = \frac{\sigma^*}{\hat{\sigma}_t} \quad (1)$$

where σ^* denotes the target volatility (10% annualized) and $\hat{\sigma}_t$ is the one-step-ahead volatility forecast generated by the estimation model at time t . When estimated volatility exceeds the target, the portfolio is scaled below full notional; when it falls below the target, exposure is scaled up. Weights are capped to prevent excessive leverage in low-volatility environments.

This base formulation is extended in the more sophisticated controller variants through the incorporation of additional risk controls, each addressing a specific failure mode of the naive scaling rule:

- **Drawdown brakes** — Programmatic reductions in exposure when cumulative losses exceed predefined thresholds, preventing the strategy from maintaining full exposure into a deteriorating drawdown.
- **Regime multipliers** — Adjustments to the base weight conditioned on the detected market regime (e.g., crisis, elevated stress, normal), enabling the strategy to distinguish between structural and transient volatility spikes.
- **Tail-risk constraints** — Hard limits on maximum portfolio weight derived from value-at-risk or expected shortfall thresholds, ensuring risk remains bounded even under model misestimation.
- **Hysteresis and no-trade bands** — Regions around the current weight within which no rebalancing occurs, reducing unnecessary turnover driven by small, noise-induced signal changes.

These enhancements are operationally critical. An unstable or poorly designed exposure mechanism can introduce discontinuous weight adjustments that complicate hedging, generate unnecessary transaction costs, and amplify portfolio volatility rather than suppressing it.

2.2 Data Description

The dataset integrates three complementary data sources, selected to provide both the raw return series required for backtesting and the rich set of features needed to train and condition advanced volatility forecasting models.

2.2.1 S&P 500 Futures — DataBento

The primary data source consists of S&P 500 futures return data obtained from DataBento, comprising both daily and intraday price records. Daily data provides the return series used in strategy backtesting and the construction of simple daily return-based volatility proxies. Intraday data—sampled at high frequency—is used to construct the robust realized volatility benchmark described below.

2.2.2 VIX Index — DataBento

The CBOE Volatility Index (VIX) is incorporated as a forward-looking implied volatility proxy. Unlike realized volatility, which is backward-looking, the VIX embeds the market’s aggregate expectation of future S&P 500 volatility over the subsequent 30-day horizon. It is included as a conditioning feature in advanced volatility forecasting models, providing a market-implied signal that is complementary to historically estimated variance.

2.2.3 U.S. Treasury Yield Curve — WRDS

Treasury yield curve data is sourced from the Wharton Research Data Services (WRDS) platform. Yield curve dynamics—including slope, curvature, and the level of interest rates—are well-documented leading indicators of macroeconomic stress and risk appetite. These series are incorporated as macroeconomic conditioning signals in the machine learning-based volatility forecasting models, enabling the models to distinguish between volatility regimes that arise from fundamentally different macroeconomic environments.

2.2.4 Realized Volatility Benchmark Construction

A central methodological contribution of this project is the construction of a robust, low-noise realized volatility benchmark using intraday data. The quality of this benchmark is of paramount importance: all volatility forecasting models are trained and evaluated against it, and a noisy or biased benchmark will produce systematically misleading model evaluations regardless of model quality.

Simple daily return-based variance estimators are susceptible to well-known biases arising from market microstructure effects—bid-ask bounce, price staleness, and asynchronous trading. These effects are most pronounced at high sampling frequencies and can cause naive realized variance estimates to be systematically upward-biased. The methodology adopted here mitigates this problem through a sub-sampling approach inspired by the work of ?.

The construction procedure is as follows:

- The intraday price series is sampled at 15-minute intervals.
- Fifteen staggered sub-grids are constructed by offsetting the sampling start point by one minute at each iteration, yielding the grids $\{1, 16, 31, \dots\}$, $\{2, 17, 32, \dots\}$, $\{3, 18, 33, \dots\}$, and so on through $\{15, 30, 45, \dots\}$.
- The realized variance is computed independently on each of the fifteen sub-grids by summing the squared intraday log-returns within each trading day:

$$RV_t^{(k)} = \sum_i r_{t,i}^{(k)2}, \quad k = 1, 2, \dots, 15 \quad (2)$$

where $r_{t,i}^{(k)}$ denotes the i -th log-return on day t within sub-grid k .

- The fifteen individual variance estimates are averaged across sub-grids, reducing the influence of any single sub-grid’s microstructure noise:

$$\overline{RV}_t = \frac{1}{15} \sum_{k=1}^{15} RV_t^{(k)} \quad (3)$$

- The averaged intraday variance estimate is annualized to obtain the daily realized volatility benchmark used throughout the project:

$$\sigma_t^{RV} = \sqrt{252 \cdot \overline{RV}_t} \quad (4)$$

This sub-sampling procedure produces a substantially more accurate and noise-robust estimate of true daily realized volatility compared to either simple daily return-based estimators or unrepeated single-grid realized variance. The resulting benchmark serves as the ground truth against which all volatility forecasting models are trained, validated, and compared in out-of-sample testing.

2.3 Evaluation Metrics

Evaluation is conducted across two complementary layers: volatility forecasting accuracy, which assesses how well each model estimates future variance; and strategy performance, which assesses how effectively each model-controller combination achieves its investment objectives when deployed within the backtesting framework. The strategy performance assessment is organized into a four-dimensional framework, ensuring that no single dimension of performance is optimized at the expense of others.

2.3.1 Volatility Forecasting Accuracy

Three metrics are used to evaluate the accuracy of volatility forecasts relative to the realized volatility benchmark.

- **Mean Squared Error (MSE)** — Measures the average squared deviation between the forecasted variance and the realized variance benchmark:

$$\text{MSE} = \frac{1}{T} \sum_{t=1}^T (\hat{\sigma}_t^2 - \sigma_t^2)^2$$

Penalizes large errors heavily due to the squared term; suitable for comparing models under symmetric loss.

- **R^2 (Out-of-Sample)** — Measures the proportion of variance in realized volatility explained by the forecasting model relative to a naive benchmark (typically the historical

mean):

$$R^2 = 1 - \frac{\sum_{t=1}^T (\sigma_t^2 - \hat{\sigma}_t^2)^2}{\sum_{t=1}^T (\sigma_t^2 - \bar{\sigma}^2)^2}$$

An out-of-sample R^2 greater than zero indicates that the model adds predictive value beyond the unconditional mean.

- **QLIKE (Quasi-Likelihood)** — A robust, asymmetric loss function particularly suited to variance forecasting in risk management contexts:

$$\text{QLIKE} = \frac{A}{P} - \ln \left(\frac{A}{P} \right) - 1$$

where A is the actual realized variance and P is the predicted variance. QLIKE penalizes underestimation of volatility more heavily than overestimation, reflecting the asymmetric cost of under-hedging in a risk management setting. A model that systematically underestimates volatility—and thereby causes the strategy to hold more exposure than is appropriate—is penalized more severely than one that conservatively overestimates it.

2.3.2 Strategy Performance: Four-Dimensional Framework

Strategy performance is assessed across four dimensions, each capturing a distinct and complementary aspect of investment quality. This multi-dimensional structure ensures that strategies are evaluated not only on their ability to generate returns, but on the full range of properties that matter to institutional investors and structured product designers.

Dimension 1 — Risk-Adjusted Returns This dimension captures the quality of returns relative to the risks taken.

- **Sharpe Ratio** — The annualized excess return of the strategy per unit of total realized volatility:

$$\text{Sharpe} = \frac{\bar{r}_p}{\hat{\sigma}_p}$$

where $\bar{r}_p$ is the annualized mean return and $\hat{\sigma}_p$ is the annualized standard deviation of returns. Higher values indicate superior risk-adjusted performance.

- **Sortino Ratio** — A refinement of the Sharpe Ratio that penalizes only downside volatility:

$$\text{Sortino} = \frac{\bar{r}_p}{\sigma_{\text{down}}}$$

The denominator uses the standard deviation of negative returns only, making it more appropriate in contexts where upside return variability is not considered harmful.

- **Calmar Ratio** — The annualized return divided by the absolute magnitude of the

maximum drawdown:

$$\text{Calmar} = \frac{\bar{r}_p}{|\text{MDD}|}$$

Captures the return generated per unit of severe downside risk; particularly relevant for drawdown-sensitive investors.

- **Total Return** — The cumulative percentage change in portfolio value over the full backtest period, measured from the initial portfolio value of \$1,000. Provides a direct, absolute measure of strategy wealth generation.

Dimension 2 — Volatility Control Accuracy This dimension evaluates the core objective of a volatility-targeting strategy: how accurately and consistently the strategy maintains realized portfolio volatility at or near the 10% annualized target.

- **Target Proximity** — The degree to which realized volatility remains close to the 10% target, measured as the mean absolute deviation of rolling realized volatility from $\sigma^* = 10\%$.
- **Volatility of Volatility (VoV)** — The standard deviation of realized rolling volatility over time. A lower VoV indicates a more stable and predictable risk profile, which is of particular value in structured product and risk-budgeting contexts.

Dimension 3 — Tail Risk and Drawdown Analysis This dimension captures how strategies behave during periods of extreme stress—precisely the conditions under which most risk management frameworks are most severely tested.

- **Maximum Drawdown** — The largest peak-to-trough percentage decline in the portfolio equity curve over the full backtest period:

$$\text{MDD} = \min_t \left(\frac{V_t - \max_{s \leq t} V_s}{\max_{s \leq t} V_s} \right)$$

Represents the worst-case cumulative loss experienced by a buy-and-hold investor in the strategy.

- **Value at Risk (VaR, 95%)** — The 5th percentile of the daily return distribution:

$$\text{VaR}_{95\%} = -\text{Quantile}_{0.05}(r_t)$$

For example, a VaR of -1.14% implies a 5% probability of losing at least 1.14% on any given trading day.

- **Days with Drawdown $> 5\%$** — The count of trading days on which the portfolio remained more than 5% below its prior all-time high. Captures not just the depth of drawdowns but their persistence, reflecting the duration of investor stress.

- **Ulcer Index** — Computed as the square root of the mean of squared daily drawdown values:

$$\text{UI} = \sqrt{\frac{1}{T} \sum_{t=1}^T D_t^2}$$

where D_t is the drawdown at time t . Penalizes both the depth and duration of drawdowns, providing a more holistic measure of cumulative portfolio stress than maximum drawdown alone.

Dimension 4 — Trading Efficiency This dimension assesses the operational efficiency of each strategy—how cleanly it converts volatility signals into executable portfolio decisions. High theoretical returns are of limited practical value if they require an impractical level of trading activity or if the payoff structure is unfavourable.

- **Turnover** — The average daily absolute change in portfolio weights:

$$\text{Turnover} = \frac{1}{T} \sum_{t=1}^T |w_t - w_{t-1}|$$

Higher turnover implies more frequent rebalancing, greater sensitivity to transaction costs, and greater operational complexity in live implementation.

- **Win-Loss Ratio** — The average return on positive-return days divided by the absolute average return on negative-return days:

$$\text{W/L} = \frac{\mathbb{E}[r_t \mid r_t > 0]}{|\mathbb{E}[r_t \mid r_t < 0]|}$$

A ratio greater than 1 indicates that gains on profitable days exceed losses on adverse days on average.

- **Win Rate** — The percentage of trading days on which the strategy generates a positive return:

$$\text{Win Rate} = \frac{1}{T} \sum_{t=1}^T \mathbf{1}(r_t > 0)$$

Considered alongside the Win-Loss Ratio to provide a complete picture of the return distribution's shape.

Collectively, these four dimensions provide a rigorous and balanced framework for comparing volatility-targeting strategies. A strategy that excels in one dimension whilst failing in another—for example, generating strong risk-adjusted returns at the cost of extreme drawdowns, or achieving low turnover at the cost of poor volatility targeting accuracy—can be identified and contextualized within this structure.

3. Volatility Estimation Models

In this section, we detail the mathematical frameworks of the volatility forecasting models implemented in this project. Beyond listing the equations, we provide the underlying motivation, key advantages, and inherent limitations of each approach to justify their selection in our volatility targeting framework.

3.1 Notation

Let r_t denote the daily log return at time t , and let $\mathcal{W}_t = \{r_{t-L+1}, \dots, r_t\}$ denote a rolling window of length L . Let $\hat{\sigma}_t$ be the volatility estimate at time t , σ^* the target annualized volatility, and $w_t^* = \frac{\sigma^*}{\hat{\sigma}_t}$ the unconstrained target weight. Unless otherwise stated, annualization is based on 252 trading days.

3.2 Volatility Estimators

The implemented volatility estimators can be broadly divided into three categories.

The first category consists of traditional historical estimators, including realized volatility and EWMA, which rely primarily on recent return dispersion.

The second category includes parametric time-series models such as AR and GARCH-family models, which explicitly model persistence, clustering, and asymmetry in volatility dynamics.

The third category consists of machine-learning and non-linear approaches, including Random Forest, LightGBM, RNN, and wavelet-based estimators, which aim to capture complex relationships and non-linear market behavior beyond classical econometric assumptions.

This categorization is important because different model families may respond differently across market regimes and therefore produce substantially different portfolio-control behavior within a volatility-targeting system.

Buy-and-Hold / Constant baseline. A trivial baseline does not forecast volatility and uses a constant reference level:

$$\hat{\sigma}_t^{\text{BH}} = \sigma^*. \quad (5)$$

Realized Volatility (RV) Using a rolling standard deviation over the last L returns:

$$\bar{r}_t = \frac{1}{L} \sum_{i=0}^{L-1} r_{t-i}, \quad \hat{\sigma}_t^{\text{RV}} = \sqrt{252} \sqrt{\frac{1}{L-1} \sum_{i=0}^{L-1} (r_{t-i} - \bar{r}_t)^2}. \quad (6)$$

- **Motivation:** To provide a non-parametric baseline that reflects recent historical price dispersion[cite: 1].

- **Advantages:** Simple to implement, transparent, and computationally efficient[cite: 1].
- **Disadvantages:** It equally weights all observations in the window, leading to "ghosting effects"[cite: 1].

EWMA The exponentially weighted variance is defined as:

$$\hat{v}_t^{\text{EWMA}} = (1 - \lambda)r_t^2 + \lambda\hat{v}_{t-1}^{\text{EWMA}}, \quad \hat{\sigma}_t^{\text{EWMA}} = \sqrt{252 \hat{v}_t^{\text{EWMA}}}. \quad (7)$$

- **Motivation:** To prioritize recent information by assigning decaying weights to past squared returns[cite: 1].
- **Advantages:** More responsive to recent shocks than simple RV and requires minimal historical data storage[cite: 1].
- **Disadvantages:** The decay factor λ is often fixed, which may not be optimal for all market conditions[cite: 1].

AR(1) Variance Model

$$\hat{v}_t^{\text{AR}(1)} = \alpha + \beta \hat{v}_{t-1}^{\text{AR}(1)}, \quad \hat{\sigma}_t^{\text{AR}(1)} = \sqrt{252 \hat{v}_t^{\text{AR}(1)}}. \quad (8)$$

- **Motivation:** To model volatility as a mean-reverting process with short-term persistence[cite: 1].
- **Advantages:** Captures the persistence of volatility by regressing tomorrow's variance on today's variance[cite: 1].
- **Disadvantages:** Assumes a linear relationship and may fail to capture high-frequency spikes[cite: 1].

AR(2) Variance Model

$$\hat{v}_t^{\text{AR}(2)} = \alpha + \beta_1 \hat{v}_{t-1}^{\text{AR}(2)} + \beta_2 \hat{v}_{t-2}^{\text{AR}(2)}, \quad \hat{\sigma}_t^{\text{AR}(2)} = \sqrt{252 \hat{v}_t^{\text{AR}(2)}}. \quad (9)$$

- **Motivation:** To incorporate additional lagged terms to capture more complex temporal dependencies[cite: 1].
- **Advantages:** Improves upon AR(1) by accounting for extended autocorrelation in variance[cite: 1].
- **Disadvantages:** Increased parameter complexity may lead to overfitting in noisy financial data[cite: 1].

GARCH(1,1)

$$\sigma_t^2 = \omega + \alpha r_{t-1}^2 + \beta \sigma_{t-1}^2, \quad \hat{\sigma}_t^{\text{GARCH}} = \sqrt{252 \sigma_t^2}. \quad (10)$$

- **Motivation:** To model volatility clustering as a function of past shocks and past variance[cite: 1].
- **Advantages:** Statistically robust for financial returns and captures the clustering phenomenon effectively[cite: 1].
- **Disadvantages:** Parameter estimation via Maximum Likelihood can be unstable in small samples[cite: 1].

GJR-GARCH

$$\hat{v}_t^{\text{GJR}} = \omega + \alpha r_{t-1}^2 + \gamma \mathbf{1}(r_{t-1} < 0) r_{t-1}^2 + \beta \hat{v}_{t-1}^{\text{GJR}}, \quad \hat{\sigma}_t^{\text{GJR}} = \sqrt{252 \hat{v}_t^{\text{GJR}}}. \quad (11)$$

- **Motivation:** To account for the empirical observation that negative shocks impact volatility more than positive ones[cite: 1].
- **Advantages:** Captures leverage asymmetry, providing more accurate risk estimates during market downturns[cite: 1].
- **Disadvantages:** More complex than standard GARCH and requires careful parameter interpretation[cite: 1].

HAR-RV

$$\hat{v}_{t+1}^{\text{HAR}} = \alpha + \beta_d RV_t^{(d)} + \beta_w RV_t^{(w)} + \beta_m RV_t^{(m)}, \quad \hat{\sigma}_{t+1}^{\text{HAR}} = \sqrt{252 \hat{v}_{t+1}^{\text{HAR}}}. \quad (12)$$

- **Motivation:** Based on the idea that traders have different investment horizons (daily, weekly, monthly)[cite: 1].
- **Advantages:** Captures long-memory properties of volatility while remaining parsimonious[cite: 1].
- **Disadvantages:** The choice of daily, weekly, and monthly horizons is subjective and fixed[cite: 1].

Lasso Volatility Model

$$\min_{\beta_0, \beta} \sum_t (r_{t+1}^2 - \beta_0 - x_t^\top \beta)^2 + \lambda \|\beta\|_1, \quad \hat{\sigma}_{t+1}^{\text{Lasso}} = \sqrt{252 \hat{v}_{t+1}^{\text{Lasso}}}. \quad (13)$$

- **Motivation:** To perform automated feature selection among many lagged volatility proxies[cite: 1].
- **Advantages:** Prevents overfitting by shrinking non-significant coefficients to zero[cite: 1].
- **Disadvantages:** Shrinkage may introduce bias, and results depend on the regularization parameter λ [cite: 1].

Random Forest

$$\hat{f}(x') = \frac{1}{B} \sum_{b=1}^B f_b(x') \quad (14)$$

- **Motivation:** To capture non-linear relationships and interactions without assuming a specific functional form[cite: 1].
- **Advantages:** Robust to outliers, prevents overfitting via bagging, and handles high-dimensional features well[cite: 1].
- **Disadvantages:** Becomes a "black-box" model, making it difficult to interpret economic drivers[cite: 1].

LightGBM

$$\hat{v}_{t+1}^{\text{LGBM}} = f_{\text{GBDT}}(x_t), \quad \hat{\sigma}_{t+1}^{\text{LGBM}} = \sqrt{252 \hat{v}_{t+1}^{\text{LGBM}}}. \quad (15)$$

- **Motivation:** To utilize gradient boosting for efficient and highly accurate non-linear forecasting[cite: 1].
- **Advantages:** High training efficiency and handles large-scale data with complex features effectively[cite: 1].
- **Disadvantages:** Can be prone to overfitting if hyperparameters are not tuned carefully[cite: 1].

RNN Volatility Model

$$h_t = \text{RNN}(X_t), \quad \hat{v}_{t+1}^{\text{RNN}} = W h_t + b, \quad \hat{\sigma}_{t+1}^{\text{RNN}} = \sqrt{252 \hat{v}_{t+1}^{\text{RNN}}}. \quad (16)$$

- **Motivation:** To leverage the sequential nature of financial data for capturing time-dependent patterns[cite: 1].
- **Advantages:** Theoretically capable of modeling complex temporal dependencies across long sequences[cite: 1].
- **Disadvantages:** Susceptible to overfitting and requires significant data and careful tuning[cite: 1].

Wavelet Volatility Estimator

$$\hat{\sigma}_t^{\text{Wavelet}} = \sqrt{252} \sqrt{\frac{1}{L-1} \sum_{i=0}^{L-1} (\tilde{r}_{t-i} - \bar{\tilde{r}}_t)^2}. \quad (17)$$

- **Motivation:** To isolate the true volatility signal by denoising returns across different frequency scales[cite: 1].

- **Advantages:** Provides a "cleaner" estimate by filtering out high-frequency noise[cite: 1].
- **Disadvantages:** Results are highly sensitive to the choice of the wavelet family and decomposition level[cite: 1].

Importantly, a volatility estimator with strong statistical forecasting accuracy does not necessarily produce the best portfolio performance once integrated into a volatility-targeting strategy.

Because portfolio weights depend directly on volatility forecasts, highly reactive estimators may generate unstable exposure adjustments, excessive turnover, or undesirable drawdown behavior. Therefore, volatility estimators are evaluated not only on forecasting accuracy, but also on how their forecasts interact with portfolio controllers in practical implementation.

4. Portfolio Control Strategies

4.1 Controllers

While volatility estimators determine the forecasted level of future risk, portfolio controllers determine how those forecasts are translated into executable portfolio exposure. In practice, the controller layer is equally important because portfolio weights directly influence leverage, turnover, drawdown behavior, and implementation stability. Even highly accurate volatility forecasts may produce poor portfolio outcomes if the exposure-adjustment mechanism reacts too aggressively or inconsistently.

The implemented controllers can be broadly categorized into static allocation rules, volatility-scaling mechanisms, regime-aware controllers, drawdown-protection systems, and tail-risk-aware exposure controls. The objective of including multiple controllers is not to identify a universally optimal exposure rule, but rather to study how different control mechanisms behave under varying market conditions and risk-management objectives.

ConstantWeight. A buy-and-hold style controller:

$$w_t = 1. \quad (18)$$

This controller serves as the unmanaged benchmark and maintains constant full exposure regardless of market volatility conditions. While simple and operationally stable, it provides no downside-risk adjustment during periods of elevated volatility.

NaiveScaling. The basic target-volatility scaling rule is

$$w_t = \frac{\sigma^*}{\hat{\sigma}_t}. \quad (19)$$

With a no-trade band δ , rebalance only when

$$|w_t^* - w_{t-1}| > \delta. \quad (20)$$

This controller represents the standard volatility-targeting mechanism. Exposure increases when forecasted volatility decreases and contracts during stressed environments. While intuitive and widely used in practice, the controller may react excessively to temporary volatility shocks and generate unstable portfolio weights.

VarianceScaling. Using inverse variance rather than inverse volatility,

$$w_t = \frac{\sigma^* \bar{\sigma}}{\hat{\sigma}_t^2}, \quad (21)$$

where $\bar{\sigma}$ is a long-run average volatility scale.

Compared with inverse-volatility scaling, this controller reacts more aggressively to changes in forecasted volatility because exposure depends on variance rather than standard deviation. This may improve responsiveness during rapid volatility increases but can also amplify turnover and leverage instability.

VolTargetClip. The clipped target-volatility rule is

$$w_t = \min\left(w_{\max}, \max\left(w_{\min}, \sigma^*/\hat{\sigma}_t\right)\right). \quad (22)$$

This controller introduces explicit upper and lower leverage constraints to prevent excessive exposure adjustments. The clipping mechanism improves practical implementability by limiting leverage expansion during extremely low-volatility environments and avoiding overly defensive deleveraging during temporary volatility spikes.

HysteresisController. Define the target weight $w_t^* = \sigma^*/\hat{\sigma}_t$. Then

$$w_t = \begin{cases} w_t^*, & \text{if } |w_t^* - w_{t-1}| > \delta, \\ w_{t-1}, & \text{otherwise.} \end{cases} \quad (23)$$

The hysteresis mechanism is designed to reduce unnecessary rebalancing caused by small fluctuations in volatility forecasts. By introducing a no-trade region around the current portfolio weight, the controller improves implementation stability and reduces turnover.

RegimeSwitchController. Let q_{low} and q_{high} denote volatility quantile thresholds. Define the regime

$$\text{Regime}_t = \begin{cases} \text{High}, & \hat{\sigma}_t \geq q_{\text{high}}, \\ \text{Low}, & \hat{\sigma}_t \leq q_{\text{low}}, \\ \text{Mid}, & \text{otherwise.} \end{cases} \quad (24)$$

Then

$$w_t = \begin{cases} \eta_{\text{high}} \frac{\sigma^*}{\hat{\sigma}_t}, & \text{High regime,} \\ \eta_{\text{low}} \frac{\sigma^*}{\hat{\sigma}_t}, & \text{Low regime,} \\ \frac{\sigma^*}{\hat{\sigma}_t}, & \text{Mid regime,} \end{cases} \quad (25)$$

where typically $\eta_{\text{high}} < 1$.

This controller explicitly incorporates market-regime information into the exposure-adjustment process. The strategy applies more defensive scaling during high-volatility regimes while allowing larger exposure during stable environments.

TrendFilter. Let $\widehat{SR}_t$ be a rolling Sharpe signal. The controller gates exposure by trend:

$$w_t = \begin{cases} \frac{\sigma^*}{\hat{\sigma}_t}, & \widehat{SR}_t \geq 0, \\ 0, & \widehat{SR}_t < 0. \end{cases} \quad (26)$$

This controller combines volatility targeting with a trend-following filter. Exposure is removed entirely during periods with negative rolling risk-adjusted returns, aiming to reduce participation during persistent downward market trends.

DrawdownBrake. Let drawdown be

$$DD_t = 1 - \frac{V_t}{\max_{\tau \leq t} V_\tau}, \quad (27)$$

where V_t is cumulative wealth. Then

$$w_t = (1 - k DD_t) \frac{\sigma^*}{\hat{\sigma}_t}, \quad (28)$$

clipped below by zero if needed.

This controller reduces exposure dynamically as cumulative portfolio losses increase. Unlike volatility-only scaling rules, drawdown-based controllers respond directly to realized portfolio deterioration and aim to prevent prolonged capital erosion during crisis periods.

DrawdownModulated. Let $d_{\min}$ and $d_{\max}$ be lower and upper drawdown thresholds. Define

$$m_t = \begin{cases} 1, & DD_t \leq d_{\min}, \\ \frac{d_{\max} - DD_t}{d_{\max} - d_{\min}}, & d_{\min} < DD_t < d_{\max}, \\ 0, & DD_t \geq d_{\max}. \end{cases} \quad (29)$$

Then

$$w_t = m_t \frac{\sigma^*}{\hat{\sigma}_t}. \quad (30)$$

Compared with the linear DrawdownBrake controller, this mechanism introduces explicit drawdown thresholds and progressively reduces exposure as portfolio losses deepen. The controller is designed to produce smoother deleveraging behavior during stressed markets.

CVaRESTargeting. Let $\widehat{ES}_t$ denote the estimated expected shortfall and ES^* the target tail risk. Then

$$w_t = \frac{ES^*}{\widehat{ES}_t}. \quad (31)$$

This controller focuses explicitly on tail-risk management by scaling exposure according to expected shortfall estimates rather than volatility alone. The objective is to improve protection against extreme downside events and severe left-tail risk.

PriorityStackController. This controller composes several risk gates multiplicatively:

$$w_t = g_t^{\text{trend}} \cdot g_t^{\text{drawdown}} \cdot g_t^{\text{tail}} \cdot \frac{\sigma^*}{\hat{\sigma}_t}, \quad (32)$$

where each gate $g_t^{(\cdot)} \in [0, 1]$ shrinks exposure according to a specific control criterion.

The controller integrates multiple layers of risk management simultaneously, including trend filtering, drawdown control, and tail-risk reduction. This structure allows the system to combine different defensive mechanisms within a unified exposure-adjustment framework.

An important insight of the framework is that portfolio performance depends not only on volatility forecasting quality, but also on how forecasts are translated into portfolio exposure. Controllers that react too aggressively may increase turnover and destabilize portfolio weights, while overly conservative controllers may fail to maintain target risk exposure efficiently. Therefore, portfolio controllers are evaluated not only on return generation, but also on implementation stability, drawdown behavior, and volatility-control accuracy.

5. Regime Detection and Modeling

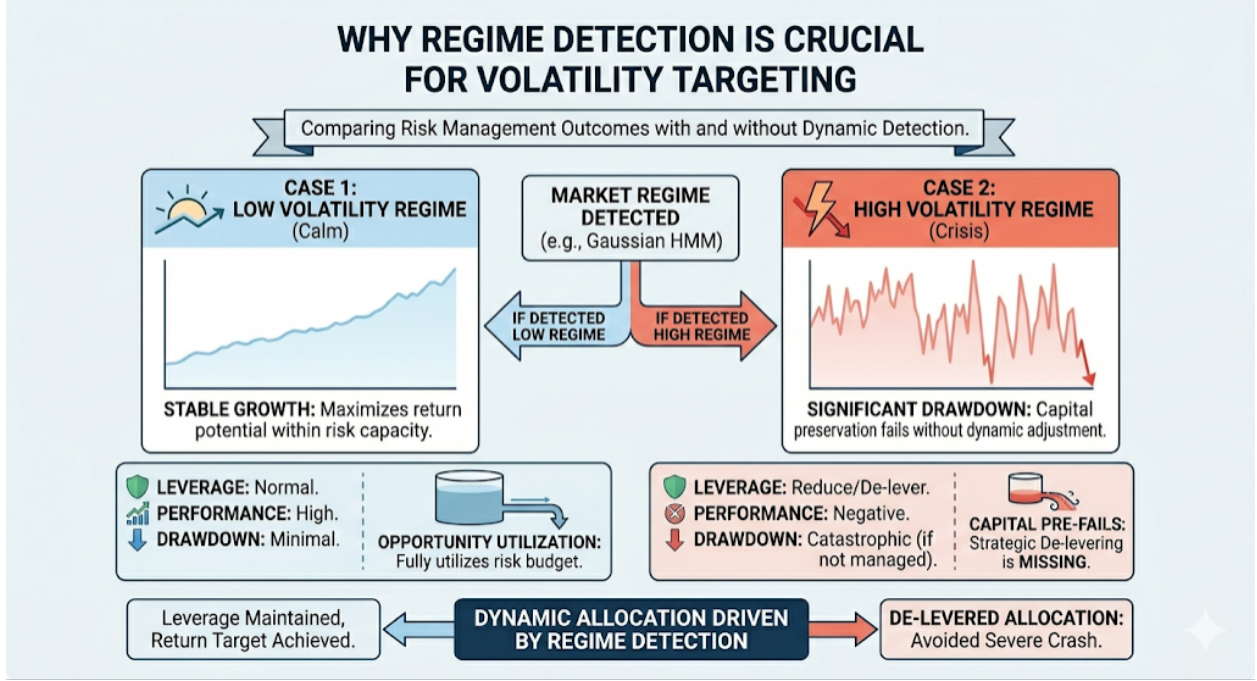


Figure 2: Regime Detection Effect

5.1 Hidden Markov Model

To capture the time-varying nature of financial market volatility, we employ a three-state Gaussian Hidden Markov Model (HMM). Volatility is not constant over time; instead, it tends to cluster into distinct periods of relative calm and extreme turbulence. The HMM provides a rigorous mathematical framework to model these unobservable, persistent market environments, referred to as latent states or market regimes.

Let $S_t \in \{1, 2, 3\}$ denote the unobserved hidden state at time t , representing low, medium, and high volatility environments, respectively. We transform the observed 20-day realized volatility (RV_{20}) of the asset by taking its natural logarithm to achieve a distribution that aligns closely with Gaussian assumptions:

$$y_t = \ln(RV_{20,t}) \quad (33)$$

The model is calibrated directly on the log realized volatility series. Conditional on being in a specific regime k at time t , the transformed observation follows a normal distribution with regime-specific mean and variance:

$$y_t \mid S_t = k \sim N(\mu_k, \sigma_k^2) \quad (34)$$

The transition between these latent states is governed by a first-order Markov chain, characterized by a transition probability matrix P , where the transition from state i to state j is defined as:

$$P_{ij} = P(S_t = j \mid S_{t-1} = i) \quad (35)$$

We optimize the parameters using the Baum-Welch algorithm to maximize the log-likelihood of the observations. Using filtered probabilities, we assign the current regime based on all data available up to time t . This allows for real-time inference of the prevailing volatility regime.

Table 1: Latent State Interpretation and Regime Characteristics

Regime Label	Latent State Interpretation	Characteristics & Dynamics
Regime 0	Low Volatility	Calm conditions, consistent upward trends, and tight distributions.
Regime 1	Medium Volatility	Consolidating markets, moderate fluctuations, rotational asset behavior.
Regime 2	High Volatility	High-stress environments, panic phases, wide dispersion of returns.

5.2 Regime Characteristics

Each identified state exhibits significantly different asset performance, risk profiles, and volatility distributions. These distinct characteristics directly influence systematic strategy implementation:

- **Volatility Behavior:** In the low volatility regime, the distribution of log-volatility has a low mean and narrow variance. Moving to the medium and high regimes, both the mean and the variance shift upward drastically, reflecting severe tail risk.
- **Risk-Return Profiles:** Strategy performance varies significantly across regimes. In Regime 0, high Sharpe ratios and stable returns allow for maximizing capital exposure. Regime 1 sees rising drawdown risk, while Regime 2 exhibits negative return expectations and heightened tail risk.
- **Motivation for Dynamic Strategy Selection:** Given the massive divergence in return distributions across different regimes, a static approach is sub-optimal. Traditional fixed-target models either take on excessive risk during market panics or leave risk capacity unutilized in quiet periods. This motivates dynamic strategy selection: altering parameters or asset weightings adaptively based on the HMM regime signal.

5.3 Alternative Regime Signals

To enhance the robustness of our regime detection, we evaluated several alternative signals. Per the current project scope, we have categorized these into signals already tested in our framework and those proposed for future research.

5.3.1 Signals Tested in Current Framework

- **Implied Volatility (VIX Index):** We have successfully integrated VIX into our testing pipeline as a forward-looking feature. Unlike realized volatility, which is backward-looking, the VIX captures market expectations of 30-day volatility. Our discussion reveals that the spread between VIX and RV_{20} often acts as a leading indicator for regime transitions. By incorporating VIX into the HMM's feature space, the model identifies the onset of "High Volatility" regimes (Regime 2) more rapidly, allowing the strategy to de-lever before severe realized drawdowns occur.

5.3.2 Proposed Signals for Future Research

While our current model focuses on price-based and implied volatility signals, the following factors are proposed as secondary filters to be explored in future iterations:

- **Macroeconomic and Credit Indicators:** Signals such as the High-Yield Credit Spread or the TED Spread provide insight into systematic liquidity risk. Future research could investigate whether these indicators help the HMM differentiate between temporary technical corrections and long-term fundamental crashes.
- **High-Frequency Intraday Metrics:** Incorporating realized variance derived from 5-minute intraday returns could potentially reduce the lag inherent in daily-frequency calculations. This would be particularly useful for identifying sudden "flash crash" scenarios that may not immediately impact daily log-realized volatility measures.

6. Combined Strategy Design

6.1 Motivation

The empirical analysis shows that no single estimator-controller pair consistently performs best across all market environments. Strategies that generate strong returns in low-volatility regimes often fail to protect against drawdowns in high-volatility periods, while defensive strategies tend to underperform during stable markets.

To address this issue, we design a regime-aware combined strategy that dynamically selects the most suitable estimator-controller pair based on the prevailing market condition. Instead of relying on a fixed strategy, the system adapts over time by conditioning decisions on an inferred latent regime. This allows different strategies to specialize in different environments, improving both robustness and risk-adjusted performance.

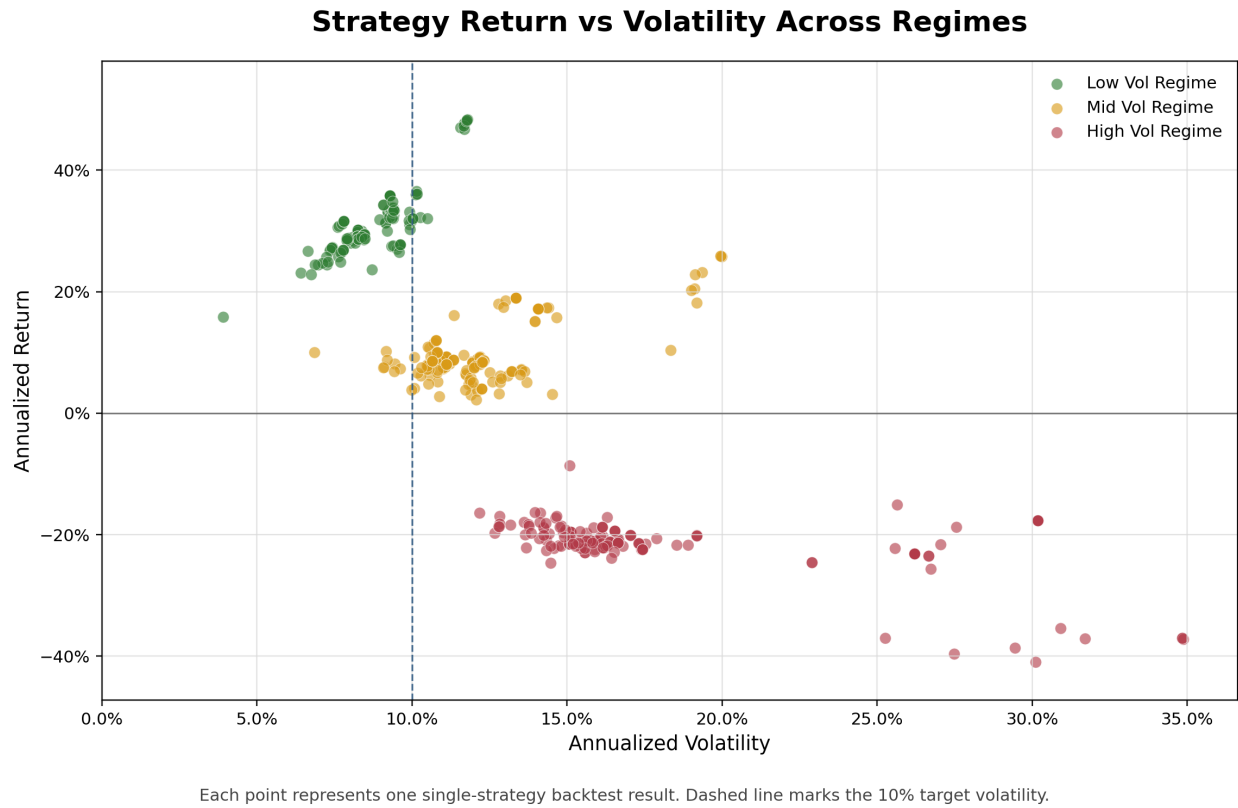


Figure 3: Strategy return versus volatility across Low, Mid, and High volatility regimes. Each point represents one single-strategy backtest result. The dashed vertical line marks the 10% annualized volatility target.

Figure 3 provides direct motivation for the combined strategy design. The main objective of the framework is not only to maximize return, Sharpe ratio, maximum drawdown, or cVaR in isolation, but to optimize these performance metrics while keeping realized volatility close

to the 10% annualized target. The figure clearly shows that strategy performance is highly dependent on the regime factor. In the Low volatility regime, many strategies are able to generate high annualized returns while maintaining volatility close to the target level. In the Mid volatility regime, returns are generally lower and more dispersed, while several strategies move further away from the target volatility. In the High volatility regime, most strategies experience negative annualized returns and substantially higher realized volatility, indicating that standalone strategies have difficulty controlling risk when market conditions deteriorate.

This regime dependence is the main reason why a fixed single strategy is insufficient. A strategy that performs well in the Low volatility regime may not remain effective in the High volatility regime, and a strategy that is defensive during stress periods may be too conservative during stable markets. Therefore, the combined strategy selects different estimator-controller pairs across regimes rather than relying on one strategy for the full sample. For example, when the objective is Sharpe ratio, the goal is to identify, within each regime, the strategy that keeps realized volatility close to 10% while producing the highest return per unit of risk. This allows the framework to pursue higher risk-adjusted performance without losing the central purpose of volatility targeting.

6.2 HMM-Based Rolling Optimization and Regime Prediction

The combined strategy is implemented within a rolling in-sample framework. At each decision date t , the engine uses a historical window $[t - N, t]$ to simultaneously infer the market regime and evaluate candidate strategies.

The regime is modeled as a latent state process and inferred using a Hidden Markov Model (HMM). The key output of the HMM is not only a classification of past regimes, but also a probabilistic estimate of the current regime based on all available historical information.

The model produces filtered state probabilities:

$$\pi_t = \mathbb{P}(S_t \mid \mathcal{F}_t),$$

where $\mathcal{F}_t$ denotes the information set up to time t . These probabilities are obtained using a forward filtering procedure, which recursively updates beliefs about the current regime as new observations arrive.

To predict the regime used for trading decisions, the engine applies the transition matrix:

$$\pi_{t+1|t} = \pi_t \cdot P,$$

where P is the estimated transition matrix of the HMM. This step projects the current filtered probabilities one step ahead, producing a forward-looking estimate of the regime at time $t + 1$.

The predicted regime is then defined as:

$$S_t = \arg \max_k \pi_{t+1|t}^{(k)},$$

which selects the most likely next-period regime. This is critical because the strategy decision at time t is applied to returns at time $t + 1$, ensuring that the framework remains strictly out-of-sample.

An important implementation detail is that the HMM transition matrix is not regenerated at every single decision date. Instead, the engine explicitly controls the refitting frequency of the HMM transition matrix and reuses the previously estimated transition dynamics between refitting dates. This design makes regime changes less noisy because the transition probabilities are not allowed to react too strongly to short-term fluctuations in the volatility signal.

This is especially important for strategy implementation. If the transition matrix were re-estimated too frequently, small temporary changes in volatility could cause the predicted regime to switch back and forth over a short time horizon. As a result, the combined strategy would frequently rotate between different estimator-controller pairs, creating unstable strategy selection and unnecessary turnover. By controlling the frequency at which the HMM regenerates the transition matrix, the engine smooths the regime prediction process and reduces short-horizon strategy shifting. Therefore, regime transitions become more persistent and economically interpretable, rather than being driven by temporary statistical noise.

6.3 Regime-Conditional Strategy Selection

Within each rolling training window, the HMM assigns historical observations to Low, Mid, and High volatility regimes. The engine then evaluates all estimator-controller combinations separately within each regime. This means that the selected strategy is not the best strategy over the full sample, but the best strategy conditional on the current market environment.

Let $R_t^{(i)}$ denote the return of strategy i , and let $w_t^{(i)}$ denote its portfolio weight. For each regime k , the engine restricts the training sample to dates classified as regime k and evaluates each strategy using only those observations.

The strategy selection is objective-specific and focuses on three criteria:

- Sharpe ratio,
- Maximum drawdown,
- Conditional Value-at-Risk (CVaR).

In addition to the raw performance metric, the engine applies two practical penalty terms. First, a turnover penalty is included to discourage strategies that require excessive changes in portfolio weights. This reflects trading costs, market frictions, and implementation instability:

$$\text{Turnover}^{(i)} = \mathbb{E} \left[|w_t^{(i)} - w_{t-1}^{(i)}| \right].$$

Second, a volatility-target mismatch penalty is applied to penalize strategies whose realized volatility deviates too far from the target volatility. Since the goal of the framework is to maintain volatility close to the target level, a strategy that produces attractive returns but fails to control realized volatility should not be selected:

$$\text{VolMismatch}^{(i)} = \left| \hat{\sigma}_{\text{realized}}^{(i)} - \sigma_{\text{target}} \right|.$$

Therefore, the adjusted objective score can be written generally as:

$$\text{AdjustedScore}_{k,m}^{(i)} = \text{PerformanceMetric}_{k,m}^{(i)} - \lambda_{\text{turnover}} \text{Turnover}_k^{(i)} - \lambda_{\text{vol}} \text{VolMismatch}_k^{(i)}.$$

Here, $\lambda_{\text{turnover}}$ and λ_{vol} represent penalty weights. They are used to incorporate implementation cost and volatility-control accuracy into the selection process, without treating turnover as a separate optimization objective.

For each regime k and objective m , the optimal strategy is selected as:

$$i^*(k, m) = \arg \max_i \text{AdjustedScore}_{k,m}^{(i)}.$$

For Sharpe ratio, the strategy with the highest adjusted score is selected. For maximum drawdown and CVaR, the same penalty logic is applied so that the engine prefers strategies with lower downside risk while still avoiding excessive turnover and poor volatility targeting.

This produces a regime-objective mapping:

$$(\text{Regime, Objective}) \rightarrow \text{Optimal Estimator-Controller Pair}.$$

As a result, the combined strategy can select different estimator-controller pairs in Low, Mid, and High regimes while still accounting for both trading efficiency and the accuracy of volatility targeting.

6.4 Out-of-Sample Execution

After the regime prediction and strategy selection are completed at time t , the engine applies the selected strategy to the next-period return.

Let S_t denote the predicted regime based on the forward probability $\pi_{t+1|t}$. The realized return is then:

$$R_{t+1}^{\text{combined},m} = R_{t+1}^{i^*(S_t,m)}.$$

This structure ensures that all decisions are made using only information available at time t , eliminating look-ahead bias.

The process is repeated sequentially:

Filter probabilities $\rightarrow$ Project regime $\rightarrow$ Select strategy $\rightarrow$ Realize next return.

The resulting return series is used to construct the equity curve:

$$\text{Equity}_t = \text{Initial Capital} \times \exp \left(\sum_{\tau \leq t} \log(1 + R_\tau^{\text{combined}}) \right).$$

6.5 Algorithm Summary

Algorithm 1 Rolling HMM-Based Combined Strategy

- 1: **for** each decision date t **do**
 - 2: Predict the volatility and assign weight using all estimator/controllers
 - 3: Construct return panel $R_t^{(i)}$.
 - 4: Define rolling training window $[t - N, t]$.
 - 5: Fit or update HMM using historical data.
 - 6: Compute filtered probabilities π_t .
 - 7: Project next-step probabilities $\pi_{t+1|t} = \pi_t P$.
 - 8: Determine predicted regime S_t .
 - 9: **for** each regime k **do**
 - 10: **for** each objective $m \in \{\text{Sharpe}, \text{MaxDD}, \text{CVaR}\}$ **do**
 - 11: Evaluate strategies within regime k .
 - 12: Apply penalty-adjusted objective.
 - 13: Select best strategy $i^*(k, m)$.
 - 14: **end for**
 - 15: **end for**
 - 16: Select strategy corresponding to (S_t, m) .
 - 17: Apply selected strategy to obtain R_{t+1} .
 - 18: **end for**
 - 19: Compute out-of-sample performance metrics.
-

6.6 Interpretation

This framework integrates regime detection and strategy selection into a unified rolling process. The HMM provides a probabilistic and forward-looking estimate of market regimes, while the transition matrix allows the model to predict future regime dynamics.

A key practical feature of the framework is that the transition matrix is not refitted at every date. Instead, the engine controls how frequently the HMM regenerates the transition

matrix. This reduces noise in regime identification because the model does not immediately reinterpret short-term volatility movements as structural regime changes. As a result, the predicted regime becomes more stable, and the selected estimator-controller pair is less likely to shift repeatedly over a short time horizon.

This stabilization is important because frequent regime switching can lead to excessive strategy rotation, higher turnover, and weaker out-of-sample robustness. By controlling the transition-matrix refitting frequency, the combined strategy keeps the regime signal adaptive while avoiding unnecessary short-term changes in strategy selection. Combined with objective-specific strategy selection, this results in a flexible and adaptive volatility targeting framework that performs robustly across different market conditions.

7. Empirical Results

7.1 Benchmark Comparison

To evaluate the effectiveness of the proposed framework, the combined strategy is compared against two benchmark strategies: buy-and-hold and naive volatility targeting.

The buy-and-hold strategy maintains full exposure to the underlying asset and therefore captures the full market return. However, this comes at the cost of significant risk exposure. As shown in Table 2, buy-and-hold achieves the highest annualized return of 7.90%, but also exhibits extremely high annualized volatility of 15.59% and a severe maximum drawdown of -36.82%. This highlights the well-known trade-off between return and risk in passive strategies.

Naive volatility targeting (22 days backward realized vol + naive scaling) improves upon this by dynamically scaling exposure based on a volatility estimate. This reduces annualized volatility to 11.50% and improves maximum drawdown to -18.41%. However, the key limitation is that this approach is *static across market regimes*. It applies the same control rule regardless of whether the market is calm or stressed, which limits its ability to adapt to structural changes in volatility dynamics.

7.2 Combined Strategy Performance

The combined strategy significantly improves risk-adjusted performance by integrating regime detection with adaptive strategy selection. Instead of relying on a single estimator-controller pair, the framework selects the best-performing combination conditional on the predicted regime.

Table 2: Performance Comparison of Combined Strategies and Benchmarks

Strategy	Sharpe	Max DD	cVaR	Ann Vol	Ann Ret
Combined (Sharpe Optimal)	0.684	-15.08%	1.65%	10.26%	6.92%
Combined (Drawdown Optimal)	0.680	-12.97%	1.55%	10.16%	6.62%
Combined (cVaR Optimal)	0.590	-14.95%	1.51%	9.74%	5.41%
Naive Volatility Targeting	0.641	-18.41%	1.85%	11.50%	6.89%
Buy-and-Hold	0.510	-36.82%	2.43%	15.59%	7.90%

Several important observations can be drawn from the results:

First, the combined strategy consistently improves the Sharpe ratio relative to both benchmarks. The best Sharpe-based combined strategy achieves a Sharpe ratio of 0.684, outperforming naive volatility targeting (0.641) and buy-and-hold (0.510). This indicates that the framework is able to generate more efficient risk-adjusted returns.

Second, the combined strategy significantly reduces downside risk. The maximum draw-down is reduced from -36.82% in buy-and-hold and -18.41% in naive volatility targeting to as low as -12.97%. Similarly, tail risk, as measured by cVaR, decreases from 2.43% (buy-and-hold) and 1.85% (naive) to 1.51% in the optimized combined strategy.

Third, the combined strategy maintains volatility close to the 10% target. This demonstrates that the framework is not only improving performance metrics but also achieving its primary objective of volatility control.

Finally, it is important to note that while buy-and-hold achieves the highest return, this comes at disproportionately higher risk. The combined strategy sacrifices a small amount of return in exchange for substantially improved stability and capital preservation.

7.3 Related Analysis

In addition to comparing the combined strategy with buy-and-hold and naive volatility targeting, we also evaluate it against the full universe of individual estimator-controller strategies. This comparison is important because the combined strategy is not designed to simply maximize raw return. Instead, its main objective is to select strategies dynamically across regimes while keeping realized volatility close to the 10% annualized target.

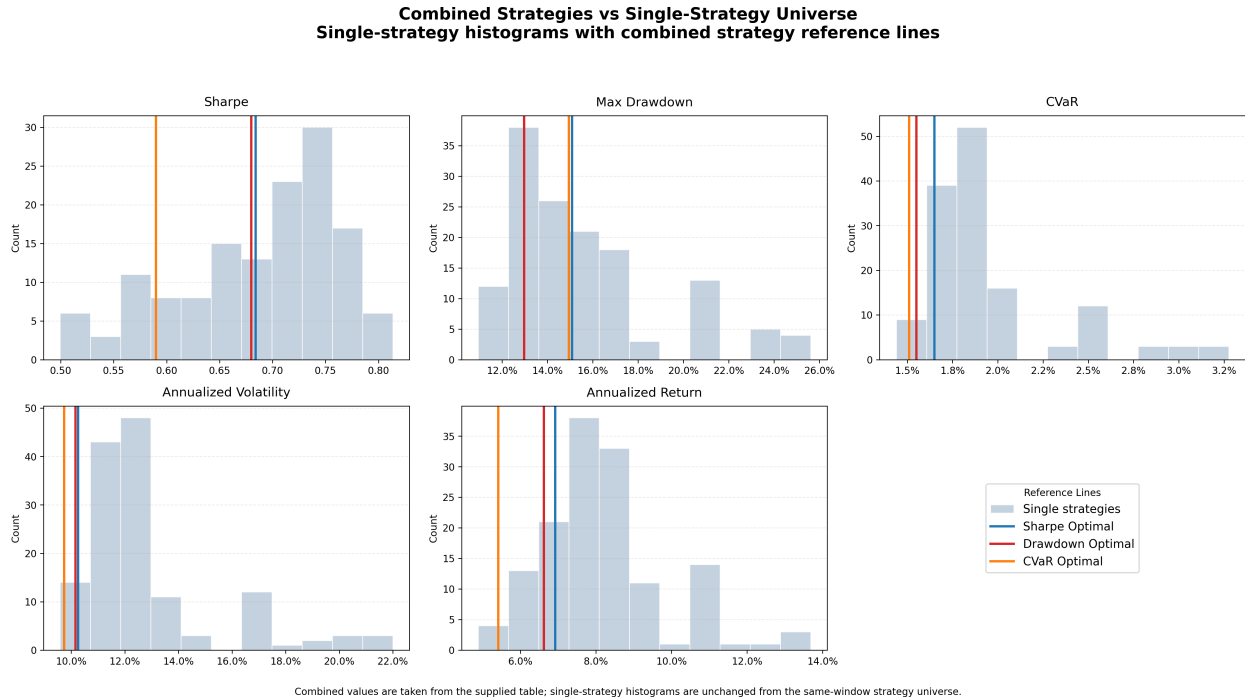


Figure 4: Combined strategies compared with the single-strategy universe. The histograms show the distribution of performance metrics across individual estimator-controller strategies, while the vertical reference lines represent the combined strategies optimized for Sharpe ratio, maximum drawdown, and cVaR.

Figure 4 compares the combined strategies with the distribution of all individual estimator-controller strategies. What we found is that a single fixed strategy is generally unable to control volatility properly across all market regimes. A strategy that performs well in a low-volatility environment may take excessive exposure when the market shifts into a high-volatility regime. Therefore, even if some individual strategies perform well historically, they do not necessarily provide stable volatility control across different regimes. The combined strategy addresses this limitation by selecting different estimator-controller pairs under different market regimes. This allows the portfolio to reduce exposure when volatility risk is elevated and maintain more efficient participation when market conditions are more stable. As a result, combining different strategies across regimes provides a more adaptive way to keep realized volatility close to the 10% annualized target.

The figure shows that the combined strategies are generally located near the left side of the annualized volatility distribution and close to the target volatility level. This confirms that the regime-based selection framework controls volatility more effectively than most standalone strategies. Many single strategies generate higher annualized returns, but they often do so by taking higher realized volatility exposure. Therefore, it is reasonable that the combined strategies do not dominate every single strategy in terms of raw return. The objective of the combined framework is not to chase the highest historical return, but to produce a more stable volatility-targeting process.

The same trade-off can also be seen in the Sharpe ratio panel. Some individual strategies achieve higher Sharpe ratios than the combined strategies, but many of those strategies also operate with higher volatility or larger downside risk. Since the combined strategy is explicitly designed to control risk exposure across regimes, a slightly lower Sharpe ratio relative to the best single strategies is acceptable. The purpose of the combined framework is not to select the best historical standalone strategy, but to reduce the instability caused by using one fixed strategy in all market environments.

The downside-risk panels provide stronger evidence for the value of the combined framework. Both the Sharpe-optimal and drawdown-optimal combined strategies achieve relatively low maximum drawdown compared with the single-strategy universe. In particular, although the Sharpe-optimal combined strategy slightly improves the Sharpe ratio, this improvement is small compared with the increase in maximum drawdown. This suggests that optimizing only for Sharpe ratio may still expose the portfolio to undesirable downside risk. By contrast, the drawdown-optimal combined strategy provides a more balanced result because it sacrifices some return-seeking behavior in exchange for stronger downside protection.

The cVaR-optimal strategy is more conservative. It produces the lowest cVaR among the combined strategies, but it also has noticeably lower Sharpe ratio and annualized return. This suggests that optimizing directly on cVaR may be too passive in this framework. Since cVaR focuses heavily on the worst tail observations, the strategy may select overly defensive combinations that reduce tail losses but also reduce participation in normal market conditions.

Therefore, cVaR is useful as a risk diagnostic, but it may not be the best primary objective for strategy selection.

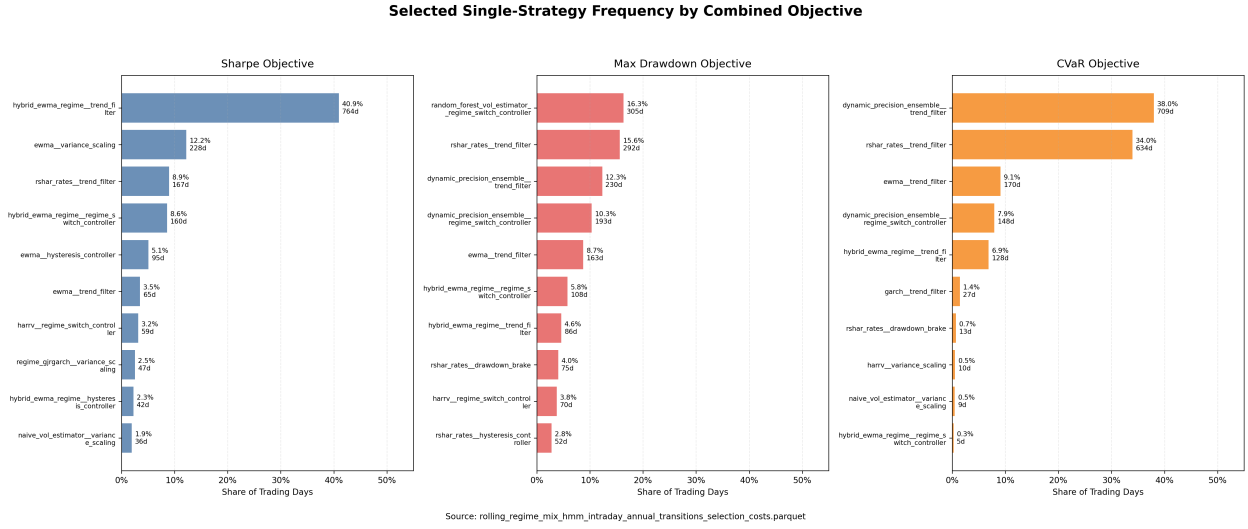


Figure 5: Selected single-strategy frequency by combined objective. Each panel shows how often each estimator-controller pair is selected by the combined strategy under the Sharpe, maximum-drawdown, and cVaR objectives. The horizontal axis reports the share of out-of-sample trading days, while the labels report both the percentage and number of selected days.

Figure 5 further explains how the combined strategy uses the single-strategy universe. The figure shows that the combined strategy does not randomly switch across all available estimator-controller pairs. Instead, each objective concentrates on a small group of strategies that are repeatedly selected over the out-of-sample period. For the Sharpe objective, the combined strategy relies heavily on the hybrid EWMA regime trend-filter strategy, which is selected on 40.9% of trading days. For the maximum-drawdown objective, the selection is more diversified across several defensive strategies, including random forest volatility estimation with a regime-switch controller, RSHAR rates with trend filtering, and dynamic precision ensemble strategies. For the cVaR objective, the selected strategies are even more concentrated in tail-risk-aware combinations, especially the dynamic precision ensemble with trend filtering and the RSHAR rates trend-filter strategy.

This selection pattern suggests that the combined strategy is flexible enough to capture the variety of the volatility targeting system, but not so noisy that it harms performance. If the strategy switched too frequently across many different single strategies, the combined framework would likely become unstable and difficult to interpret. However, the observed selection frequencies show that the engine mainly rotates among a limited set of strong candidates. Therefore, the combined strategy can still adapt to different volatility environments while maintaining stable strategy selection. This supports the idea that the HMM-based regime prediction and penalty-adjusted selection process provide enough flexibility to respond

to market changes, but enough stability to avoid excessive short-horizon noise.

7.4 Why the Combined Strategy Works

The superior performance of the combined strategy can be explained by several structural advantages.

First, regime adaptation. Financial markets exhibit regime-dependent behavior, especially in volatility clustering. A single model cannot capture all dynamics. By conditioning on regimes, the strategy effectively decomposes the problem into simpler sub-problems.

Second, model diversification. The framework evaluates a large number of estimator-controller combinations. Different models capture different aspects of volatility (e.g., persistence, jumps, mean reversion). The combined strategy acts as a dynamic ensemble, selecting the most suitable model in each regime.

Third, forward-looking regime prediction. The use of HMM transition probabilities allows the strategy to anticipate regime changes rather than react to them. This reduces lag and improves responsiveness during market stress.

Finally, risk-focused objective design. By explicitly optimizing for drawdown and tail risk in certain regimes, the strategy aligns more closely with practical portfolio management objectives, rather than purely maximizing return.

7.5 Key Findings

The empirical results confirm that regime-aware volatility targeting provides substantial benefits. Forecast accuracy alone is not sufficient to guarantee strong trading performance; instead, the interaction between volatility estimation and portfolio control plays a crucial role.

The combined strategy demonstrates that adaptive selection across regimes can significantly improve risk-adjusted returns, reduce drawdowns, and stabilize portfolio performance. This supports the central hypothesis of the project: incorporating regime information into both forecasting and control layers leads to a more robust and effective volatility targeting system.

8. Recommendations and Potential Impacts

8.1 Practical Recommendations

The empirical results of this project suggest that effective volatility targeting should be implemented as a regime-aware system rather than as a single static strategy. Different estimator-controller combinations perform differently across market environments, and no individual strategy consistently dominates in all volatility regimes. The combined strategy framework improves robustness by dynamically adapting strategy selection according to the inferred market state.

The results show that the combined strategies significantly reduce downside risk relative to both buy-and-hold and naive volatility targeting benchmarks. Maximum drawdown is reduced from -36.82% under buy-and-hold and -18.41% under naive volatility targeting to as low as -12.97% in the combined framework, while maintaining competitive annualized returns and improved Sharpe ratios. These findings suggest that institutional volatility-control systems should incorporate regime-adaptive allocation mechanisms rather than relying on a fixed estimator or controller.

Another important finding is that forecasting accuracy alone is insufficient for evaluating volatility-targeting strategies. Models with strong statistical forecasting performance do not necessarily produce superior portfolio outcomes once transaction costs, turnover, and drawdown behavior are considered. The interaction between volatility estimation and portfolio control plays a critical role in overall strategy quality. Therefore, evaluation should extend beyond forecasting metrics such as QLIKE and MSE to include implementation-focused metrics such as turnover, volatility-target adherence, and downside-risk stability.

The results also highlight the importance of balancing return generation against stability and capital preservation. Although buy-and-hold achieves the highest annualized return, it also exhibits substantially higher volatility and significantly larger drawdowns. The combined strategy sacrifices a small amount of return in exchange for materially improved downside protection and more stable volatility control. For institutional portfolios and structured products, this trade-off is often preferable because maintaining stable risk exposure is typically more important than maximizing raw return.

8.2 Potential Business Impact

The framework developed in this project has several potential applications in institutional portfolio management and structured product design.

First, the framework can support volatility-controlled structured products and systematic overlay strategies. More stable volatility targeting reduces abrupt leverage adjustments and improves the consistency of portfolio exposure, which can simplify hedging operations and reduce hedge instability during stressed market environments.

Second, the modular architecture makes the framework scalable and operationally flexible. New volatility estimators, controllers, or regime signals can be incorporated without redesigning the entire system. This flexibility is valuable for institutions seeking to adapt risk-management systems across different products or asset classes.

Third, the regime-aware selection process improves practical implementability by incorporating turnover penalties and volatility-target mismatch penalties directly into the optimization framework. This helps reduce excessive trading activity and improves robustness under realistic transaction-cost assumptions.

Finally, the project demonstrates that adaptive volatility targeting can provide more stable risk-adjusted performance across different market environments. This may improve portfolio stability, reduce severe drawdowns, and enhance long-term capital preservation for institutional investors operating under strict risk-management constraints.

9. Conclusion and Future Work

9.1 Conclusion

This project develops and evaluates a regime-aware volatility-targeting framework that integrates volatility forecasting, portfolio control, and dynamic strategy selection within a unified backtesting system.

The empirical results confirm that volatility-targeting performance is highly regime-dependent. No single estimator-controller pair consistently dominates across all market conditions. Instead, adaptive strategy selection conditioned on market regimes produces more stable and robust portfolio outcomes.

Relative to buy-and-hold, the combined strategy substantially improves downside protection while maintaining competitive returns. Maximum drawdown is reduced from -36.82% to as low as -12.97% , while the Sharpe ratio improves from 0.510 to 0.684. Compared with naive volatility targeting, the combined framework also achieves stronger tail-risk control and more stable volatility targeting performance.

An important conclusion of the project is that forecasting accuracy alone is insufficient for determining overall strategy quality. The interaction between volatility estimation and portfolio control plays a critical role in determining realized portfolio behavior. Strategies with strong statistical forecasting performance may still produce unstable exposure adjustments, excessive turnover, or poor downside-risk characteristics.

The primary contribution of this project is therefore not a single forecasting model, but rather a modular and reproducible framework for systematically evaluating volatility estimators, portfolio controllers, and regime-aware allocation mechanisms under consistent out-of-sample testing conditions. The framework demonstrates that incorporating regime information into both forecasting and control layers can improve robustness, downside protection, and overall portfolio stability.

9.2 Limitations

Several limitations should be acknowledged.

First, the analysis is conducted only on S&P 500 futures. Although this provides a clean and liquid testing environment, the results may not generalize directly to other asset classes such as fixed income, commodities, or cryptocurrencies.

Second, transaction costs are modeled using simplified assumptions. Real-world implementation frictions, particularly during periods of market stress, may be higher than those assumed in the backtesting framework.

Third, the Gaussian Hidden Markov Model assumes relatively stable transition dynamics and Gaussian state distributions. In practice, financial markets may experience abrupt structural breaks or extreme tail events that are not fully captured by the model.

Finally, several machine-learning-based estimators are sensitive to hyperparameter selection, training-window specification, and feature construction. Model stability may therefore vary across different market environments and sample periods.

9.3 Future Work

Several directions may extend the current framework.

First, future research may expand the framework to multi-asset portfolios, including fixed income, commodities, and cryptocurrencies. This would allow the study of cross-asset volatility transmission and regime spillover effects.

Second, additional macroeconomic and credit-market indicators may be incorporated into the regime-detection framework. Variables such as credit spreads, liquidity measures, and macroeconomic indicators may improve the early identification of structural market stress.

Third, future work may explore online and adaptive learning methods in which volatility models and regime-transition probabilities update continuously over time. This may improve responsiveness to changing market dynamics while preserving stability in strategy selection.

Finally, the framework may be extended beyond discrete strategy selection toward dynamic ensemble weighting approaches. Instead of selecting a single estimator-controller pair within each regime, future systems may allocate capital across multiple strategies simultaneously based on probabilistic regime information and real-time performance diagnostics.

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